Irene (Hui-Ru) Ho | Curriculum Vitae — *October 2026*

PhD Student · Department of Computer Sciences

University of Wisconsin-Madison, 1205 University Ave, Madison, WI 53726

Email: hho24@wisc.edu

Homepage: https://irene-huiru-ho.github.io

Google Scholar: https://scholar.google.com/citations?user=OiUSjXYAAAAJ&hl=en

LinkedIn: www.linkedin.com/in/irene-hui-ru-ho/

Education

2024 - present PhD Computer Sciences, University of Wisconsin–Madison, Madison, WI, USA
- Advisor: Dr. Bilge Mutlu (People and Robots Laboratory)
- GPA: 3.9/4.0

2021 - 2023 MS Computer Sciences, University of Wisconsin–Madison, Madison, WI, USA
- Advisor: Dr. Bilge Mutlu

2019 - 2023 MS Educational Psychology, University of Wisconsin–Madison, Madison, WI, USA
- Advisor: Dr. Edward Hubbard (Educational Neuroscience Laboratory)

2014 - 2019 BS Psychology, National Taiwan University , Taipei, Taiwan
- Minor in Economics

Research Experience

09/2019 - Present *Graduate Researcher*, **People and Robotics Lab, UW-Madison**, Madison, WI
- Advisor: Prof. Bilge Mutlu

06/2026 - 10/2026 *Applied Scientist Intern II*, **Fauna Robotics, Amazon**, Sunnyvale, CA
- Advisors: Dylan Glas, Theo Benig and Xiaoxu Liu
- Project: Developed Interaction Strategies for Household Robots' Long-horizon Task Planning and Execution

09/2025 - 03/2026 *Visiting Researcher*, **HAI Lab, Singapore Management University**, Singapore
- Advisor: Prof. Jiannan Li
- Project: Investigated AI-powered robotic manipulation and generative UI could enhance human learning.

2019-2023 *Graduate Researcher*, **Educational Neuroscience Lab, UW–Madison**, Madison, WI
- Advisor: Prof. Edward Hubbard
- Project: Investigated how attention and multisensory perception affect working memory and numerical cognition.

Summer 2022 *User Experience Researcher Intern – AI Platform Team*, **Meta Platforms, Inc.**, Menlo Park, CA
- Led mixed-methods studies collaborating with AI/ML engineers to support AI productivity and debugging experience.

Summer 2017 *User Experience Researcher Intern – Consumer Team*, **Trend Micro Taiwan**, Taipei, Taiwan
- Led mixed-methods research to develop privacy-sensitive products and evaluated in-home parental control device.

Publications

† **Mentored undergraduate student**

Full Conference Paper

Under Review — **Ho, H. R.**, Xu, M., Kujava, N., Mutlu, B. (Under Review). Understanding Physical Safety Awareness of Embodied AI for Household Robots. *Manuscript*

Ho, H. R., Mutlu, B. (Under Review). Designing Embodied AI for Informal Learning: How a Mobile Robot Can Support In Situ Discovery in the Physical World. *Manuscript*

2026 — Rahman, M. A., Hatch, P., **Ho, H. R.**, Felix, I. A., Mutlu, B., Michaelis, J. E. HomeGuide: Empowering Teachers with a Generative AI Authoring System for Personalized Robot-Assisted Learning. Proceedings of the ACM Symposium on User Interface Software and Technology (UIST '26) *Acceptance Rate: 24%*

Ho, H. R., Cagiltay, B., Wang, J., Ibtasar, R., Mutlu, B., Michaelis, J. Exploring Student Perspectives on Interacting with Social Robots for Homework. Proceedings of the 22nd Annual ACM Interaction Design and Children Conference (IDC '26) *Acceptance Rate: 24%*

Sato, A., Kim, C., White, N., Maneesh, A., Wang, Y., **Ho, H. R.**, Mutlu, B. NarrativeBlocks: Generative Scaffolding in End-User Design and Programming of Social Robots. Proceedings of the 2026 ACM Conference on Designing Interactive Systems (DIS '26). *Acceptance Rate: 21%*

† Liu, Z., † Wang, Y. K., **Ho, H. R.**, Wu, Y. H., Zhao, Y. H., Mutlu, B. AskNow: An LLM-powered Interactive System for Real-Time Question Answering in Large-Scale Classrooms. Proceedings of the 2026 CHI conference on human factors in computing systems (CHI '26). *Acceptance Rate: 25%*

2025 — **Ho, H. R.**, Kargeti, N., Liu, Z., Mutlu, B. SET-PAiREd: Designing for Parental Involvement in Learning with an AI-Assisted Educational Robot. Proceedings of the 2025 CHI conference on human factors in computing systems (CHI '25). *Acceptance Rate: 24%*

Kim, C., Sato, A., White, N., **Ho, H. R.**, Lee, C., Huang, Y., Mutlu, B. Bridging Generations using AI-Supported Co-Creative Activities. Proceedings of the 2025 CHI conference on human factors in computing systems (CHI '25). *Acceptance Rate: 24%* **Honorable Mention Award (Top 251 in 5014)**

2024 — **Ho, H. R.**, Hubbard, E. M., Mutlu, B. "It's Not a Replacement:" Enabling Parent-Robot Collaboration to Support In-Home Learning Experiences of Young Children. Proceedings of the 2024 CHI conference on human factors in computing systems (CHI '24). *Acceptance Rate: 25%* **Honorable Mention Award (Top 150 in 4028)**

2023 — **Ho, H. R.**, White, N. T., Hubbard, E. M., Mutlu, B. Designing Parent-child-robot Interactions to Facilitate In-Home Parental Math Talk with Young Children. In Proceedings of the 22nd Annual ACM Interaction Design and Children Conference (IDC '23) (pp. 355-366). *Acceptance Rate: 29%*

2021 — **Ho, H. -R.**, Cagiltay, B., White, N., Hubbard, E., Mutlu, B. RoboMath: Designing a Learning Companion Robot to Support Children's Numerical Skills. ACM Interaction Design and Children Conference (IDC '21), Athens, Greece. *Acceptance Rate: 36%*

2020 — Cagiltay, B., **Ho, H. -R.**, Michaelis, J., Mutlu, B. Investigating Family Perceptions and Design Preferences for an In Home Robot. ACM Interaction Design and Children Conference (IDC '20), London, United Kingdom. *Acceptance Rate: 32%*

Short Paper & Extended Abstracts

Under Review — **Ho, H. R.**, Benig, T., Liu, X. X., Glas, D. F. Developing Interaction Strategies for Teaching a Robot Household Tasks *Manuscript*

2026 — **Ho, H. R.** Empowering Human Learning through Human-Robot Interaction. In Companion of the 2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI '26). *Acceptance Rate: 23%*

2025 — **Ho, H. R.** Empowering Parents and Teachers to Support Children's Learning through AI-based and Robotic Learning Companions. Extended Abstract of the 2025 CHI conference on human factors in computing systems (CHI EA '25). *Acceptance Rate: 18%*

Awards & Honors

March 2026	*HRI '26 Pioneers (Acceptance rate: 23%)*, HRI 2026	$1300
April 2025	*CHI '25 Best Paper Honorable Mention Award (Top 251 in 5014)*, CHI 2025	
April 2025	*CHI '25 Doctoral Consortium Award (Acceptance rate: 18%)*, CHI 2025	$1800
May 2024	*CHI '24 Best Paper Honorable Mention Award (Top 150 in 4028)*, CHI 2024	
June 2023	*Graduate Student Travel Award*, University of Wisconsin-Madison	$500
June 2023	*Student Research Grants Competition*, University of Wisconsin-Madison	$1500
May 2022	*Government Scholarship to Study Abroad (GSSA)*, Ministry of Education, Taiwan (R.O.C.)	$32,000
July 2018	*Fully-Funded Scholarship: Study Abroad Program for Future Scholars in the Humanities and Social Sciences*, Ministry of Education, Taiwan (R.O.C.)	$30,000

Mentoring Experience

2026-	Nathan Kujava, Undergraduate Assistant	Madison, WI
2025-	Sophia Chen, Undergraduate Assistant (Honors)	Madison, WI
2026	Grace Luo, Undergraduate Assistant–*Current CS MS at Univ. of California-San Diego*	Madison, WI
2025-2026	Sungwoon Park, Undergraduate Assistant–*Current Software Enigneer at ServiceNow*	Madison, WI
2025-2026	Arushi Dodhia, Undergraduate Assistant–*Current Robotics MS at Santa Clara Univ.*	Madison, WI
2024-2025	Tony (Yuankun) Wang, Undergraduate Assistant	Madison, WI
2023-2025	Ziqi Liu, Undergraduate Assistant (Honors)	Madison, WI
2023-2025	Justina Wang, Undergraduate Assistant (Honors)–*Current CS PhD at Northwestern Univ.*	Madison, WI
2023-2025	Nitigya Kargeti, Graduate Assistant	Madison, WI
2024	Yash Sancheti, Undergraduate Assistant–*Current Software Engineer at Microsoft*	Madison, WI
2022-2024	Muhammad Junaid, Graduate Assistant–*Current Software Developer at Whova*	Madison, WI

Teaching Experience

SU 2025	*Teaching Assistant*, CS 540 Intro to Artificial Intelligence	UW-Madison
SU 2024	*Instructor*, UW-Madison Grandparents University	UW-Madison
2022 - 2024	*Teaching Assistant*, EP 301 How People Learn	UW-Madison
2020 - 2022	*Teaching Assistant*, EP 326 Mind, Brain and Education	UW-Madison

Service and Outreach

March 2027	*Lead Program Chair*, ACM/IEEE HRI 2027 Pioneers Workshop	Santa Clara, CA
April 2026	*Student Host and Coordinator*, Academic Seminar by Prof. Jodi Forlizzi from CMU HCII	Madison, WI
April 2026	*Student Volunteer*, ACM CHI 2026	Barcelona
March 2025	*Student Host and Coordinator*, Academic Seminar by Prof. Ying Xu from HGSE	Madison, WI
May 2024	*Student Volunteer*, ACM CHI 2024	Honolulu, HI
June 2023	*Student Volunteer*, ACM IDC 2023	Chicago, IL

Peer Review

CHI	2024*, 2025*, 2026*, 2027
HRI	2025, 2026, 2027
IDC	2022, 2023, 2024, 2026
IUI	2026
DIS	2026
ICHEC	2026

* *Special recognition for outstanding reviews*